

## MSE 311 Thermodynamics of materials-Homework 2

Name:

Student ID:

*Please hand in this homework before class on September 22<sup>th</sup>.*

1. Consider how to kill coronavirus by using heat.
2. Try to find the first paper which mentions Calvin knowing 0 K equals  $-273.15^{\circ}\text{C}$ , and write down the title and the published year.

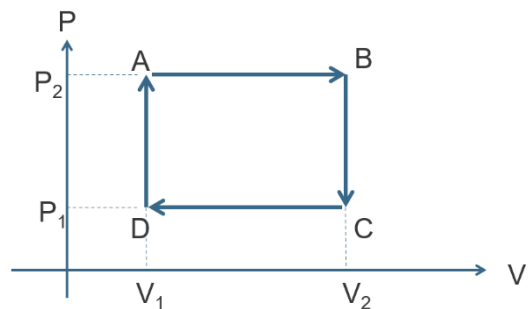
3. One mole of an ideal gas are taken round the following cycle. Please find the efficiency of the cycle.

A  $\rightarrow$  B: Constant Pressure Process

B  $\rightarrow$  C: Constant Volume Process

C  $\rightarrow$  D: Constant Pressure Process

D  $\rightarrow$  A: Constant Volume Process



Open question. If there is no temperature scale, how to derive the Carnot efficiency;  
*Drinking bird* Perpetual motion principle.